



UNIVERSIDAD DE BUENOS AIRES
FACULTAD DE CIENCIAS EXACTAS Y NATURALES
DEPARTAMENTO DE COMPUTACIÓN

[your title here]

Tesis de Licenciatura en Ciencias de la Computación

Rafael Romani

Director: Santiago Cifuentes

Codirector: Santiago Figueira

Buenos Aires, 2026

[YOUR TITLE HERE]

[Acá va el resumen]

Palabras claves: [acá van las palabras clave]

[YOUR TITLE HERE]

[write your abstract here]

Keywords: [write your keywords here]

[acá van los agradecimientos]

CONTENTS

1. Introduction	1
1.1 Uso y poder de los LLMS	1
1.2 Pero qué es un LLM?	2
1.3 Formalizaciones previas y resultados que ignoran los aspectos probabilísticos	2
1.4 Objetivos y contribuciones	2
2. Theoretical Framework	3
2.1 Transformers	3
2.2 Finite precision modeling	6
2.3 Transformer Complexity classes	7
2.4 Circuit Complexity	8
2.5 Known results	8
3. General properties	9
3.1 Transformers Primitives	9
3.2 Normalization	16
3.3 Boolean Operations	17
4. Main results	22
5. Conclusions	23

1. INTRODUCTION

1.1 Uso y poder de los LLMS

In 2017, a group of researchers working at Google published a paper introducing a technology that revolutionized society. In *Attention is All You Need*, they presented the transformers, today's core of large language models (LLMs) and generative AI.

This architecture was build intended as text sequence-to-sequence model for machine translation but today has applications in various fields as natural language processing, computer vision or speech recognition.

The first model implementing transformers open for public use was ChatGPT by OpenAI. It made its debut in November 2022 and by December 2024 it reported 300 million global weekly users. Now there are many models competing for this market as Claude, from Anthropic, or Gemini from Google.

These models can be all be grouped under the generative AI tools. This is a technology that is arguably the most rapid technology adopted by society. In the second half of 2025, roughly one in six people worldwide were using generative AI tools. not sure how to cite this <https://www.microsoft.com/en-us/research/wp-content/uploads/2026/01/Microsoft-AI-Diffusion-Report-2025-H2.pdf>

LLMs are being used not only as chatbots for asking small questions, but are shaping the industry and future. For example, [?] found that more than 17% of the Computer Science's papers published between January 2020 and February 2024 on the *arXiv*, *bioRxiv*, and *Nature* portfolio journals had sentences heavily edited by ChatGPT.

Missing usage in industry

So naturally, the question about their computing power pops up. Are this kind of models really capable of doing the stuff for what they are being used?

- well-suited to parallelism enables transformer models to take full advantage of the power and speed offered by GPUs during both training and inference. This possibility, in turn, unlocked the opportunity to train transformer models on unprecedentedly massive datasets through self-supervised learning.
- chatgpt, claude, gemini,
- every day users
- number of users?
- use in the industry?
- other uses besides the chatbots?

SF: Lo citás como miscellaneous

cite this <https://www.ibm.com/model>

1.2 Pero qué es un LLM?

1.3 Formalizaciones previas y resultados que ignoran los aspectos probabilísticos

1.4 Objetivos y contribuciones

2. THEORETICAL FRAMEWORK

2.1 Transformers

In this work we are going to analyze transformers as language recognizers. A transformer is a machine that works over an alphabet Σ . It has only one tape from where it reads the input and in an autoregressive

SF no sé bien a qué se refiere con autoregressive, pero tiene que quedar claro que solo imprime nuevos símbolos de a uno y que no puede modificar símbolos ya escritos. Quizá comparar con autómata vs TM

way writes in it more tokens. There are two distinguished tokens called **decision symbols**. Once the transformer writes one of those, it halts. We say that a transformer accepts a word if, when it halts, the decision symbol printed is the accepting one (q_{accept}). Analogously, a transformer rejects a word if the decision symbol printed is the rejecting one (q_{reject}).

In each step, the transformer maps the content of the tape to a probability distribution over Σ . Depending on this distribution, the transformer prints the next token. The transformer iterates this process until it halts. These iterations are called **chain of thought**.

The process of generating the next token consists of two separated steps. First a probability distribution over the alphabet for the next token is computed by a process called distribution generation.

SF así como hiciste con cot, definí nuevos términos con un estilo especial. Yo uso defstlye como comando y después decido si es itálica, negrita, etc

RR defino distribution generation y token selection más abajo y los puse en negrita. Debería marcarlos también acá?

After that, it is necessary to decide the specific token by a process called token selection. These two algorithms together is what defines the transformer. During this work we will analyze and compare two different options for token selection.

In most of the literature, the transformers operate over an infinite tape. This is far from the real models, thus the transformers studied in this thesis work over a finite tape. The size of the tape is defined, for each input size n , as $\text{budget}(n) + n$. It is important to remark that the size of the tape bounds the number of steps of CoT that the transformer makes. For an input word of length n , no computation can take more than $\text{budget}(n)$ steps because every step write a token in the tape.

SF mejorar esta última oración, no se entiende

RR ahí va mejor?

The process of distribution generation is dependent on many parameters. In the real models, these parameters are the ones calculated during the training.

SF Revisar casos como estos: These parameters, in the real models, are the ones calculated during the training. → In the real models, these parameters are the ones calculated during the training.

We refer to them as θ and consider them as part of the definition of a transformer.

We note $\text{TF}(\alpha)$ to the token outputted by one step of the transformer TF when the tape content is $\alpha \in \Sigma^*$. The autoregressive token generation is defined recursively. The token

SF: los pasos son del cómputo, model transformer. Revisar en todo el texto

outputted after i steps is $\text{TF}^i(\alpha) := \text{TF}^{i-1}(\alpha \text{TF}(\alpha))$ with base case $\text{TF}^1(\alpha) := \text{TF}(\alpha)$. In other words, the output of the i th step is $\sigma_i = \text{TF}^i(\alpha) = \text{TF}(\alpha \sigma_1 \dots \sigma_{i-1})$ where $\sigma_j \in \Sigma$ for all j .

We note $\text{TF}^*(\alpha)$ for the token written by the transformer when it halts, i.e. at the end of the computation, when the input is α . As said before, the transformer only halts when it prints a decision symbol (q_{accept} or q_{reject}). Therefore, $\text{TF}^*(\alpha) \in \{q_{\text{accept}}, q_{\text{reject}}\}$.

As stated before, the **distribution generation** is a mapping of the content of the tape to a probability distribution over Σ . There are different algorithms for this process in the literature. All the transformers considered in this work use the one presented by [?]. This algorithm consists of four parts: a token embedding layer (TE), a position encoding layer (PE), an output linear layer (OUTPUT) and a stack of L identical decoder layers, where L is also called the depth of the model. Each decoder layer has two sub-layers: a multi-head self-attention layer (ATTN) and a position-wise fully-connected feed-forward network (FF). Each part and layer has its own trainable parameters. Then, we can define the distribution generation algorithm by its parameters $\theta = \{\theta_{\text{TE}}, \theta_{\text{PE}}, \{\theta_{\text{ATTN}}^l, \theta_{\text{FF}}^l\}_{l=0}^{L-1}, \theta_{\text{OUTPUT}}\}$. The algorithm works over vectors of numbers of certain dimension d . The dimension is dependent on the size of the input word. We call $d(n)$ to the **embedding size** when the input word has size n .

Token Embedding: It is a mapping from Σ to vectors in $\mathbb{R}^d$. This mapping is defined by $\theta_{\text{TE}} \in \mathbb{R}^{d \times |\Sigma|}$. For all $\sigma \in \Sigma$, we note $\theta_{\text{TE}}(\sigma) \in \mathbb{R}^d$ to the mapping of token σ .

Position Encoding: It is a mapping from the positions of the tape to vectors in $\mathbb{R}^d$. This mapping is defined by $\theta_{\text{PE}} \in \mathbb{R}^{d \times \text{budget}(n)+n}$. For all position of the tape $1 \leq i \leq \text{budget}(n) + n$ we note $\theta_{\text{PE}}(i) \in \mathbb{R}^d$ to the mapping of position i .

Self Attention Layer: It is a mapping from $(\mathbb{R}^d)^a$ to $(\mathbb{R}^d)^a$. Given parameters $\theta_{\text{ATTN}} = (W_Q, W_K, W_V, W_O) \in \mathbb{R}^{d \times d} \times \mathbb{R}^{d \times d} \times \mathbb{R}^{d \times d} \times \mathbb{R}^{d \times d}$, the layer is defined by the algorithm 1. This layer is defined only as a single head attention layer because we can simulate 1 multi-head attention layer with any constantly many heads with multiple single-head attention layers[?].

Algorithm 1 Causal Self-Attention, ATTN

Input: Parameter $\theta_{\text{ATTN}} = (W_Q, W_K, W_V, W_O)$ and embedding $h = (h_1, \dots, h_a) \in \mathbb{R}^d \times \dots \times \mathbb{R}^d$.

Output: Embedding $h' = (h'_1, \dots, h'_a) := \text{ATTN}_{\theta_{\text{ATTN}}}(h_1, \dots, h_a)$.

- 1: $q_i := W_Q h_i, k_i := W_K h_i, v_i := W_V h_i, \forall i \in [1, \dots, a]$
 - 2: $s_i := \text{softmax}(\langle q_i, k_1 \rangle, \dots, \langle q_i, k_i \rangle) \parallel (0, \dots, 0)$
 - 3: $h'_i := W_O \sum_{j=1}^a (s_i)_j v_j$
-

Feed Forward Network: It is a mapping from $\mathbb{R}^d$ to $\mathbb{R}^d$. Given $\theta_{\text{FF}} = (W_1, b_1, W_2, b_2) \in \mathbb{R}^{d \times d} \times \mathbb{R}^d \times \mathbb{R}^{d \times d} \times \mathbb{R}^d$, the layer is defined as $\text{FF}_{\theta_{\text{FF}}}(h) := W_2 \text{relu}(W_1 h + b_1) + b_2$, where $\text{relu} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ replaces each position x_i by $\max(0, x_i)$.

SF Too many it's it's... Cambié algunos. Mejorar

Output Layer: It is a mapping from $\mathbb{R}^d$ to a probability distribution over Σ . Given $\theta_{\text{OUTPUT}} \in \mathbb{R}^{|\Sigma| \times d}$, we define it as $\text{OUTPUT}_{\theta_{\text{OUTPUT}}}(h) := \text{softmax}(\theta_{\text{OUTPUT}} h)$.

All these parts come together for the distribution generator process in the algorithm 2.

SF: of reals?

RR: son $\mathbb{R}_{e,s}$. Puedo decir simplemente que son números por ahora?

SF: ¿qué es a ?

RR: es la cantidad de elementos que hay en la cinta en ese paso. Hay un vector por cada elemento. Cada paso de ATTN consume todos los vectores y devuelve la misma cantidad de vectores. En el algoritmo aparece a . No se me ocurrió otra forma para notarlo.

Algorithm 2 Distribution generator

Input: Transformer parameter $\theta = (\theta_{\text{TE}}, \theta_{\text{PE}}, \{\theta_{\text{ATTN}}^l, \theta_{\text{FF}}^l\}_{l=0}^{L-1}, \theta_{\text{OUTPUT}})$ and tokens of the tape $\alpha \in \Sigma^*$.

Output: distribution $p_\theta(\cdot|\alpha)$.

- 1: $h_i^{(0)} \leftarrow \theta_{\text{TE}}(\alpha_i) + \theta_{\text{PE}}(i), \forall 1 \leq i \leq |\alpha|$
- 2: **for** $l = 0, \dots, L - 1$ **do**
- 3: $(h_1^{(l+0.5)}, \dots, h_{|\alpha|}^{(l+0.5)}) \leftarrow (h_1^{(l)}, \dots, h_{|\alpha|}^{(l)}) + \text{ATTN}_{\theta_{\text{ATTN}}^l}(h_1^{(l)}, \dots, h_{|\alpha|}^{(l)})$
- 4: $h_i^{(l+1)} \leftarrow h_i^{(l+0.5)} + \text{FF}_{\theta_{\text{FF}}^l}(h_i^{(l+0.5)}), \forall 1 \leq i \leq |\alpha|$
- 5: **end for**
- 6: $p_\theta(\cdot|\alpha) \leftarrow \text{OUTPUT}_{\theta_{\text{OUTPUT}}}(h_{|\alpha|}^{(L)})$

The OUTPUT layer and the ATTN layer use a softmax function $\mathbb{R}^n$ to $\mathbb{R}^n$ defined as $\text{softmax}(x)_i = \exp(x_i) / \sum_{j=1}^n \exp(x_j)$.

SF Ejemplo de cómo mejorar la presentación para repetir menos y ser más claro:

The OUTPUT layer and the ATTN layer uses a softmax function. It is defined from $\mathbb{R}^n$ to $\mathbb{R}^n$ for all n as $\text{softmax}(x)_i = \exp(x_i) / \sum_{j=1}^n \exp(x_j)$.

→

The OUTPUT layer and the ATTN layer use a function $\mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $\text{softmax}(x)_i = \exp(x_i) / \sum_{j=1}^n \exp(x_j)$.

Independientemente de la presentación, no sé si queda claro el tema del n ("for all n "). ¿Es una familia de funciones softmax_n ?

RR Creo que formalmente es una familia de funciones porque es distinta para cada tamaño de vector, pero en el fondo hace siempre lo mismo. Quizás borrando el para todo n queda más claro? No creo que valga la pena introducir que es una familia.

The second component of a transformer is the **token selector**. This algorithm consumes the probability distribution computed by the distribution generator and returns a token.

Let $p : \Sigma \rightarrow [0, 1]$ be the output of the distribution generator. In this work we study two selectors, we call them **deterministic selector** and **probabilistic selector**. The deterministic selector always returns the token with the largest probability (argmax over p). The probabilistic selector works non-deterministically. It returns a token $\sigma \in \Sigma$ with probability $p(\sigma)$.

RR Add an image representing the transformer architecture

Definition 2.1.1 (Deterministic transformer). We call a transformer deterministic when the token selector is the deterministic selector. We say that a family of deterministic transformers $\{\text{TF}_n\}_{n \in \mathbb{N}}$ recognizes a language $\mathcal{L}$ when:

$$\alpha \in \mathcal{L} \iff \text{TF}_{|\alpha|}^*(\alpha) = q_{\text{accept}}$$

As mentioned in chapter 1, there is a theoretical gap in the area. A study of non-determinism in transformers is missing.

When TF is a deterministic transformer, $\text{TF}^i(\alpha)$ is an exact symbol for all $\alpha \in \Sigma^*$ and $i \in \mathbb{N}$. On the contrary, when the token selection process of the transformer samples the distribution, it becomes a random variable. Therefore, we introduce the following definition.

SF: Σ^* ?

RR: No, el token selector consume la distribución que dado un token devuelve su probabilidad

RR: está bien dicho theoretical gap?

SF: no, theoretical gap no me suena bien acá. ¿A qué te referís?

RR: a que hay

Definition 2.1.2 (Probabilistic transformer). We call probabilistic transformer to the transformers that use the probabilistic selector. We say that a family of probabilistic transformers $\{\text{TF}_n\}_{n \in \mathbb{N}}$ recognizes a language $\mathcal{L}$ when:

$$\alpha \in \mathcal{L} \iff \Pr[\text{TF}_{|\alpha|}^*(\alpha) = q_{\text{accept}}] > 2/3$$

SF ¿Dónde quedó el budget en estas definiciones?

RR El budget solo se usa para acotar la cantidad de pasos de CoT. También aparece en el position embedding para poder argumentar que es una función que hace cualquier cosa pero con dominio finito. Si el dominio no fuera finito se rompe la demo de que puedes simular un paso de cómputo con un circuito.

2.2 Finite precision modeling

I'm missing good symbols for ∞ y $-\infty$.

In practice, training and inference of transformers are typically done with 16- or 32-bit floating point numbers and there is active work in reducing that number. That's why in this thesis we follow the decision of [?] to work on constant-precision transformers, where the output of each arithmetic operation is rounded to the closest floating point number representable by a fixed number of digits following IEEE 754 standard. This analysis avoids the unrealistic infinite precision assumption made in prior works [?]. The definitions stated in the section are inherited from [?].

SF:
gramática

SF The definitions stated in this section are taken from [?].

Below we give a formal definition of the floating-point number and rounding operation. We use $\phi(a) = \sum_{i=1}^k 2^{k-i} a_i$ to denote the value of the binary number represented by $a \in \{0, 1\}^k$ for any $k \in \mathbb{N}^+$.

SF Antes de la definición, explicar cuál es la representación que se usa, qué es el exponent y el significand. ¿Es base y exponente? Y de sing... Poner un ejemplito

Definition 2.2.1 (Floating-point Representation). Let e be the number of bits for exponents and s be the number of bits for significand. A $(e + 2s + 1)$ -bit binary string $a = (a_1, a_2, \dots, a_{e+2s+1}) \in \{0, 1\}^{e+2s+1}$ is a floating-point binary representation of number $\phi_{e,s}(a) \triangleq \text{sign}(a) \cdot 2^{\text{exponent}(a)} \cdot \text{significand}(a)$ with e -bit exponent and $2s$ -precision, where the sign is $\text{sign}(a) \triangleq 2a_1 - 1$, the significand is $\text{significand}(a) \triangleq 2^{-s} \phi(a_{2:2s+1})$, and the exponent is $\text{exponent}(a) \triangleq \phi(a_{2s+2:2s+e+1}) - 2^{\max(0, e-1)}$. We define the set of numbers expressible with e bits of exponent and s bits of significand as $\mathbb{F}_{e,s}$. We define $\infty_{e,s}$ as the maximum number in $\mathbb{F}_{e,s}$. Similarly, we define $-\infty_{e,s}$ as the minimum. Since in this work we only analyze constant precision transformers, we will omit the subscripts of ∞ and $-\infty$.

Definition 2.2.2 (Correct Rounding). For any $x \in \mathbb{R}$, we define correct rounding $\text{round}(x, \mathbb{F}_{e,s})$ as the closest number to x in $\mathbb{F}_{e,s}$. We break the tie by picking the one with a smaller absolute value. In particular, we denote the rounding operation with e -bit exponent, $2s$ -bit precision by $\text{round}_{e,s}(\cdot) \triangleq \text{round}(\cdot, \mathbb{F}_{e,s})$, which is also denoted by $[\cdot]_{e,s}$ for convenience. We extend the definition of round and $\text{round}_{e,s}$ to vector inputs by rounding coordinate-wisely.

Our notion of floating-point number simplifies the IEEE 754 Standard for Floating-point Arithmetic [?] by removing ∞ and $-\infty$. When overflow happens, we always round the output to ∞ , and when underflow happens to $-\infty$. For unary functions like $\exp(\cdot)$ and

binary functions including addition, subtraction, multiplication, and division, we simply define their rounded version by rounding their outputs. Whenever division by 0 happens, we define the output as 0.¹

Next, we define finite-precision summation over more two numbers by decomposing it as a chain of rounded binary addition in a fixed order.²

SF:
gramática

Definition 2.2.3 (Summation with Iterative Rounding). For any $s, n \in \mathbb{N}^+$ and vector $x \in \mathbb{R}^n$, we define summation with iterative rounding to e bit exponent and $2s$ -bit precision as $\text{sum}_{e,s} : \cup_{n \in \mathbb{N}^+} (\mathbb{F}_{e,s})^n \rightarrow \mathbb{F}_{e,s}$, where for any $n \in \mathbb{N}^+$ and $x \in \mathbb{R}^n$,

$$\text{sum}_{e,s}(x) \triangleq \left[\left[\left[[x_1 + x_2]_{e,s} + x_3 \right]_{e,s} + \cdots + x_{n-1} \right]_{e,s} + x_n \right]_{e,s}.$$

We further define the following operations:

- Finite-precision inner product: $\langle x, y \rangle_{e,s} \triangleq \text{sum}_{e,s}(x \odot y)$;
- Finite-precision matrix product: $(A \times_{e,s} B)_{i,j} \triangleq \langle (A_{i,:})^\top, B_{:,j} \rangle_{e,s}$;
- Finite-precision softmax: $\text{softmax}_{e,s}(x) \triangleq \left[[\exp(x)]_{e,s} / \text{sum}_{e,s}([\exp(x)]_{e,s}) \right]_{e,s}$.

From the saturation and the iterative rounding presented by [?], some properties appear that are important to remark:

- $\exp(-\infty) = 0$
- $\exp(\infty) = \infty$
- $-\infty \cdot \infty = -\infty$
- $\infty \cdot \infty = \infty$
- $\forall x \in \mathbb{F}_{e,s}$ it holds that $x + 0 = x$
- $\forall x \in \mathbb{F}_{e,s}$ it holds that $x - x = 0$
- $\forall x \in \mathbb{F}_{e,s}$ it holds that $x + \infty + \infty + -\infty = 0$

2.3 Transformer Complexity classes

Maybe rethink this title?

As discussed in the first chapter, there are different resources to bound in the transformers to limit their expressive power. In this thesis, we analyze the effect of bounding the number of steps in the chain of thought that the transformers are allowed to make before answering. With this in mind, we define the following complexity classes.

Given two functions $T(n)$ and $d(n)$ we define the complexity class $\text{CoT}[T(n), d(n)]$. Informally, a language $\mathcal{L}$ is in $\text{CoT}[T(n), d(n)]$ if and only if there is a deterministic family of transformers with constant depth, embedding size $d(n)$ and budget $T(n)$ that recognizes it.

¹ Notice that the only division is in the softmax. If it's the one in the attention mechanism, it's reasonable to give an attention score of 0 to every vector. If it's the one in the output layer it means that all symbols have an extremely small probability, therefore is reasonable to assign the same probability to all of them.

² Technically speaking, instead of a chain, the summation could also proceed like a tree.

Definition 2.3.1 ($\text{CoT}[T(n), d(n)]$). A language $\mathcal{L}$ is in $\text{CoT}[T(n), d(n)]$ if and only if there is an integer L and two functions $T'(n) = O(T(n))$, $d'(n) = O(d(n))$, such that for every positive integer n , there is an L -layer deterministic transformer, denoted by TF_n with embedding size $d'(n)$, that outputs $\mathcal{L}(\alpha)$ given any input α in Σ^n , using $T'(n)$ steps of chain of thought.

SF Explicar qué es $\mathcal{L}(\alpha)$.

RR No es notación estándar?

In the same way, we define the classes of languages recognized by probabilistic transformers. Given two functions $T(n)$ and $d(n)$ we define the complexity class $\text{RCoT}[T(n), d(n)]$. Informally, a language $\mathcal{L}$ is in $\text{RCoT}[T(n), d(n)]$ if and only if there is a probabilistic family of transformers with constant depth, embedding size $d(n)$ and budget $T(n)$ that recognizes it.

Definition 2.3.2 ($\text{RCoT}[T(n), d(n)]$). A language $\mathcal{L}$ is in $\text{RCoT}[T(n), d(n)]$ if and only if there is an integer L and two functions $T'(n) = O(T(n))$, $d'(n) = O(d(n))$, such that for every positive integer n , there is an L -layer probabilistic transformer, denoted by TF_n with embedding size $d'(n)$, that output $\mathcal{L}(\alpha)$ given any input α in Σ^n , using $T'(n)$ steps in the chain of thought with probability higher than $2/3$.

2.4 Circuit Complexity

2.5 Known results

3. GENERAL PROPERTIES

In this chapter we study the properties of $\text{CoT}[T(n), d(n)]$ classes and the primitive operations that transformers can perform. We define the notion of normalized transformer and prove that $\text{CoT}[T(n), d(n)]$ classes are closed under boolean operations.

SF: mejorar la redacción

3.1 Transformers Primitives

SF En esta sección se presentan las primitivas que se van a usar a lo largo de la tesis. Estas primitivas son operaciones que pueden ser implementadas por un transformer con ciertas características.... bla bla. Motivar y explicar por qué se necesita esta sección. Mencionar cuáles van a ser las primitivas y decir que la explicación de cada una de ellas se encuentra en las subsecciones correspondientes.

RR: ahí va mejor?

Flagging Tokens and Positions

For complex operations inside the loop of the transformer we will need to identify when a vector comes from a certain token or position. In this section we present how to mark the vectors with flags during the embedding.

For flagging a token we use the token embedding, and in the same way, for flagging a position we use the position encoding. Either way, the strategy is exactly the same. The embedding dimension is extended to allocate the flags. A new coordinate is added per flag. Therefore, using a constant number of flags only adds a constant to the embedding dimension. Thus, if the flagging is made in a family of transformers its $\text{CoT}[T(n), d(n)]$ class does not change.

For example, let v_σ and v_τ be vectors of a transformer TF where v_σ comes from embedding the token σ and v_τ from the token τ (with $\sigma \neq \tau$). We can define a transformer TF' that behaves like TF but recognize when a vector comes from embedding the token σ . TF' have embedding dimension $d' = d + 1$ where d is the embedding of TF .

$$v_\sigma = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \rightarrow v'_\sigma = \begin{pmatrix} x \\ y \\ z \\ f_{\text{on}} \end{pmatrix} \quad v_\tau = \begin{pmatrix} i \\ j \\ k \end{pmatrix} \rightarrow v'_\tau = \begin{pmatrix} i \\ j \\ k \\ f_{\text{off}} \end{pmatrix}$$

Fig. 3.1: Change in a flagged and unflagged vector

Formally, for flagging the token σ we define the token embedding function of TF' : $\Sigma \rightarrow \mathbb{R}^{d+1}$ as

$$(\theta'_{TE}(\tau))_j = \begin{cases} (\theta_{TE}(\tau))_j & \text{if } j \neq d+1 \\ f_{\text{on}} & \text{if } j = d+1 \text{ and } \tau = \sigma \\ f_{\text{off}} & \text{if } j = d+1 \text{ and } \tau \neq \sigma \end{cases}$$

where f_{on} is the value that marks the vector as flagged and f_{off} as unflagged. Those values can be defined arbitrary for each flag. In this case, the coordinate inserted for the flag is

RR: arbitrary no es la palabra que quiero me parece

the last one, but it can be added anywhere.

The other components of TF' ignore $d + 1$ the coordinate. The position encoding of TF' is equal to the position encoding of TF , except it puts a 0 in the $d + 1$ coordinate (not present in TF) independently of the position. The matrix of the ATTN , FF and OUTPUT layers of TF' are the ones of TF extended with 0 in the $d + 1$ rows and columns to match the change in embedding dimension.

TF' behaves exactly as TF except it has one more coordinate in the embedding. Said coordinate represent the flag and in every vector has only values v_{on} or v_{off} .

This primitive works also as a mechanism to extend the embedding dimension without changing the behavior. If v_{on} and v_{off} are defined as 0, the resulting transformer is the same as the initial with an extra coordinate that it is always 0.

Preserve and ignore coordinates through ATTN and FF

When defining new transformers over existing ones we will need to extend the embedding dimension to store new information. For preserving the behavior of the original transformers through the attention and feed forward layers, we need to build a mechanism that ignores the extra coordinates.

Let TF be a transformer of embedding dimension d and assume we are constructing a transformer TF' of embedding dimension $d' = d + 1$. Let W_Q, W_K, W_V, W_O be the matrix of an ATTN layer of TF and assume we want to perform the same layer in TF' but simulating that the i th coordinate does not exist. Notice that the embedding dimension of TF' has one more coordinate than TF 's. Defining the matrix of the layer W'_Q, W'_K, W'_V, W'_O as W_Q, W_K, W_V, W_O but inserting a row and column of zeros in the i th positions gives the desired ATTN layer. Note that because the i th row of W'_O is zero, the output vectors of the ATTN layer are 0 in the i th coordinate. This is the point that makes it possible to not just ignore the value of the i th coordinate, but to not affect it.

For preserving the coordinate but not ignore it, it is enough to have zeros in only its corresponding row of W'_O . This construction works for more than one coordinate. For extending it to a set of coordinates, it is only needed to make more rows and columns 0.

The same applies in the feed forward layer, making the i th column and row of the matrix and vectors of the layer zero, makes it ignore the value of the i th coordinate. Also in the result of the FF will appear a zero in the i th coordinate, thus also preserving the value.

Add vector and constant functions

In this section we develop a mechanism for adding a fixed vector to a previously flagged one. This technique works even when there are more than one flagged vector.

Assume there is a flag with values $f_{\text{on}} = 1$ and $f_{\text{off}} = 0$. For clarity, suppose the flag is in the last coordinate. Let v be the vector we want to add. We define a mechanism that for all h such that $h_d = f_{\text{on}}$, it replaces them with $h + v$.

The operation is performed with only one FF layer. We construct a layer such that $\text{FF}(h) = \mathbb{I}(h_d = f_{\text{on}}) \cdot v$. Said layer is achieved with the following parameters.

$$W_1 = id \quad W_2 = \begin{pmatrix} 0 & \dots & 0 & v_1 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & 0 & v_d \end{pmatrix} \quad b_1 = b_2 = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

Notice that if the coordinate for the flag had been the k th instead of the last one, then v would have appeared in the k th column of W_2 instead of the last.

This technique is that it can be used to apply a constant function to the flagged vectors. This is achieved exploiting the finite representation system. We can saturate the vector to the ∞ and then make it 0 to finally add our objective. This holds since for all $x, c \in \mathbb{F}_{e,s}$ it happens that $x + \infty + \infty + -\infty + c = c$. Because as mentioned in section 2.2, $x + \infty + \infty = \infty$ and $\infty + -\infty = 0$.

The constant function $f(h) = v$ is performed adding four vectors such that performs this trick. First, it is necessary to add twice the vector with only ∞ , then add a vector with only $-\infty$ and finally add v . As is visually presented in equation 3.1.

$$h \leftarrow h + \begin{pmatrix} \infty \\ \vdots \\ \infty \end{pmatrix} + \begin{pmatrix} \infty \\ \vdots \\ \infty \end{pmatrix} + \begin{pmatrix} -\infty \\ \vdots \\ -\infty \end{pmatrix} + v$$

Fig. 3.2: Constant function

RR más arriba digo ecuación pero latex dice que es una figura. Tampoco estoy seguro qué caption ponerle. Quizás es mejor si directamente no la referencio?

Make a vector ignore others in an ATTN layer

We will find very useful to be able for some vectors to ignore others in the ATTN layers.

SF Otro ejemplo de redacción rara:

"We will find very useful to be able for some vectors to ignore others in the ATTN layers."

→

We show how to make some vectors ignore others in the ATTN layers, which will be very useful for our constructions.

For a vector to ignore other, we will need for the attention score to be $-\infty$ so the softmax make it zero.

Let's say we want for the l th vector to ignore a set of vectors M . Therefore, what we are looking for is for $\langle q_l, k_m \rangle = -\infty$ when $m \in M$, but we don't want to modify the attention score of others pairs. We will use the saturation error for getting to our objective, since $\langle q_l, k_m \rangle = \sum_{x=1}^d (q_l)_x (k_m)_x$ if we make for the last two terms to be $-\infty$ when $m \in M$, we will get what we are looking for. Since we don't want to affect other vectors, every other pair should have 0 as the last two terms.

We will make $(k_i)_{d-1}$ and $(k_i)_d$ to be $-\infty$ when $i \in M$, and $(q_i)_{d-1}$ and $(q_i)_d$ to be ∞ when $i = l$ and 0 otherwise. With these values the only attention scores modified are the ones of the l th vector since every other will have 0 in the last two coordinates of the query thus making the last two terms of the internal product 0 and therefore not affecting the score.

For getting those values in the keys and queries vectors we can just flag them and adjust the query and key matrix to have what we want in those coordinates.

SF: parece muy informal ("to have what we want"). ¿Hay una mejor manera de de-

Linear transformations

Although it looks that transformers are a model that should be able to perform linear transformations in a native way, it is not the case. Applying a linear transformation to a vector is not direct since in every step of ATTN and FF the result of the layer is added to the original vector instead of replacing it. A sharp reader maybe would think that this can be solved applying the transformation $M - I$ when we want to transform the vector x to Mx , since $x + (M - I)x = x + Mx - Ix = Mx$. This does not hold in our model since the addition and multiplication in the finite representation of the numbers used is not commutative. For avoiding the representation error, we will extend the embedding dimension to get more memory per vector such that we can store in new coordinates the value of Mx and then replace it in the original ones. After applying this process we will have corrupted all vectors except the one we are interested in.

SF: corrupted?

The strategy we will use is the following. For having a clearer notation, we will say that the vector to which we are applying the linear transformation is the f th and the matrix of the linear transformation is M . We will do use two consecutive ATTN layers where the f th vector will only attend to itself. We can achieve this making it ignore all the others vectors except itself, then its attention score to itself will be 1.

SF: Podés no dar bola, pero f no es un buen nombre para una coordenada. ¿Qué tal i, j, k, r, s, u, v ?

The first layer will compute the result of the transformation and store it in new coordinates added to the embedding to be used as memory. The second layer will move the result to the right place.

For doing this we will need to extend the embedding dimension with enough coordinates to store the result of the transformation. Therefore, if we are working with a transformer of embedding dimension d and want to do a linear transformation of an entire vector, we will need to extend the dimension by a factor of 2. The first d coordinates will be for the original vector and the second d coordinates will be the ones using as memory. We will assume that the d coordinates used as memory are set to 0 by the layers previous to this mechanism. This can be achieved maintaining them as 0 since the beginning or transformed to 0 with the primitive described in the previous section.

$$h_f = \begin{pmatrix} x_1 \\ \vdots \\ x_d \\ 0 \\ \vdots \\ 0 \end{pmatrix} \xrightarrow{\text{first ATTN layer}} h_f = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ (Mx)_1 \\ \vdots \\ (Mx)_d \end{pmatrix} \xrightarrow{\text{second ATTN layer}} h_f = \begin{pmatrix} (Mx)_1 \\ \vdots \\ (Mx)_d \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

this should be an epigraph: a graphic representation of the technique

Let's construct each of the ATTN layers. As said before, in both layers we will need for h_f to only attend to itself. Therefore, we define W_Q and W_K using the strategy of the previous section.

Let's remember from the definition of the ATTN mechanism (1) that the output of the ATTN layer is $h'_i = W_O \sum_{j=1}^a (s_i)_j v_j$. Let's analyze what h'_f looks by now:

$$h'_f = W_O \sum_{j=1}^a (s_f)_j v_j = W_O v_f = W_O (W_V h_f)$$

SF: dividir claramente en primera y segunda layer. Podés usar itemize o paragraph.

Then, defining W_O and W_V as follows

$$W_V = id \qquad W_O = \begin{pmatrix} -id^{\mathbb{R}^{d \times d}} & 0^{\mathbb{R}^{d \times d}} \\ M & 0^{\mathbb{R}^{d \times d}} \end{pmatrix}$$

makes us get $h'_f = (-x_1, \dots, -x_d, (Mx)_1, \dots, (Mx)_d, 1)$, which added to h_f gives us what we were looking for after the first layer.

For the second ATTN layer, we can use the same mechanism for making h_f only attend to itself. And defining W_V and W_O as follows

$$W_V = id \qquad W_O = \begin{pmatrix} 0^{\mathbb{R}^{d \times d}} & id^{\mathbb{R}^{d \times d}} \\ 0^{\mathbb{R}^{d \times d}} & -id^{\mathbb{R}^{d \times d}} \end{pmatrix}$$

we get $h'_f = ((Mx)_1, \dots, -(Mx)_d, -(Mx)_1, \dots, -(Mx)_d)$, which added to h_f gives us what we were looking for after the second layer.

Max coordinate

We will see later on that we will need to be able to take the maximum of a set of coordinates.

SF "we will see later on that we will need to be able to" es demasiado recargado. Podés decir "We show how to take the maximum of a set of coordinates, which will be useful for our constructions."

The operation we are looking to do is the following:

$$\begin{pmatrix} a \\ b \end{pmatrix} \rightarrow \begin{cases} \begin{pmatrix} a \\ 0 \end{pmatrix} & \text{if } a > b \\ \begin{pmatrix} 0 \\ b \end{pmatrix} & \text{if } b > a \end{cases}$$

We will define it for taking the maximum between only two coordinates, but it can be extended to a larger set repeating the operation as a tournament.

This operation can be done with the following primitives:

$$\begin{aligned}
& \begin{pmatrix} a \\ b \\ 0 \\ 0 \\ \vdots \end{pmatrix} \xrightarrow{\text{FF layer}} \begin{pmatrix} a \\ b \\ \text{relu}(a-b) \\ \text{relu}(b-a) \\ \vdots \end{pmatrix} \xrightarrow[\text{the coordinates with the relu}]{\text{Multiply twice by } \infty} \begin{pmatrix} a \\ b \\ \infty \cdot \mathbb{I}(a > b) \\ \infty \cdot \mathbb{I}(b > a) \\ \vdots \end{pmatrix} \rightarrow \\
& \xrightarrow[\text{coordinates with the comparison}]{\text{Divide by } \infty \text{ the}} \begin{pmatrix} a \\ b \\ \mathbb{I}(a > b) \\ \mathbb{I}(b > a) \\ \vdots \end{pmatrix} \xrightarrow[\text{operation}]{\text{Special}} \begin{pmatrix} a \cdot \mathbb{I}(a > b) \\ b \cdot \mathbb{I}(b > a) \\ 0 \\ 0 \\ \vdots \end{pmatrix} = \begin{cases} \begin{pmatrix} a \\ 0 \\ \vdots \end{pmatrix} & \text{if } a > b \\ \begin{pmatrix} 0 \\ b \\ \vdots \end{pmatrix} & \text{if } b > a \end{cases}
\end{aligned}$$

SF Ok, pero esto no da exactamente lo que se buscaba, ¿no?

Implementation of the special operation

The operation we are looking for to do it's not a linear transformation, therefore we need to define a new technique. Remember that one of $\mathbb{I}_a$ and $\mathbb{I}_b$ is a 1 and the other is a 0. Notice that $\mathbb{I}_b = 1 - \mathbb{I}_a$ and $\mathbb{I}_a = 1 - \mathbb{I}_b$.

$$\begin{pmatrix} a \\ b \\ \mathbb{I}_a \\ \mathbb{I}_b \end{pmatrix} \xrightarrow[\text{operation}]{\text{Special}} \begin{pmatrix} a \cdot \mathbb{I}_a \\ b \cdot \mathbb{I}_b \\ 0 \\ 0 \end{pmatrix}$$

SF:
gramática.
Pasar
revisor.
No
marco
nada de
redacción
o
gramática

This operation can be performed with three layers of FF. What we will do is the following:

$$\begin{pmatrix} a \\ b \\ \mathbb{I}_a \\ \mathbb{I}_b \end{pmatrix} \xrightarrow[\text{operation}]{\text{Special}} \begin{pmatrix} a + \infty \cdot \mathbb{I}_b + \infty \cdot \mathbb{I}_b + -\infty \cdot \mathbb{I}_b \\ b + \infty \cdot \mathbb{I}_a + \infty \cdot \mathbb{I}_a + -\infty \cdot \mathbb{I}_a \\ \mathbb{I}_a + 0 + 0 + -\mathbb{I}_a \\ \mathbb{I}_b + 0 + 0 + -\mathbb{I}_b \end{pmatrix}$$

If $\mathbb{I}_a = 1$, then $\mathbb{I}_b = 0$, so the first coordinate isn't modified and the second is changed to 0 because of the finite representation properties. In the same way, if $\mathbb{I}_a = 0$, then $\mathbb{I}_b = 1$, so the second coordinate isn't modified and the first is changed to 0.

As mentioned before, this operation is performed with three FF layers. The first two will be:

$$W_1 = id \quad W_2 = \begin{pmatrix} 0 & 0 & 0 & \infty \\ 0 & 0 & \infty & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad b_1 = b_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Notice that the output of this FF layer when inputted with $(a, b, \mathbb{I}_a, \mathbb{I}_b)$ will be $(\infty \cdot \mathbb{I}_b, \infty \cdot \mathbb{I}_a, 0, 0)$.

The third FF layer will be:

$$W_1 = id \quad W_2 = \begin{pmatrix} 0 & 0 & 0 & -\infty \\ 0 & 0 & -\infty & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad b_1 = b_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

which inputted with $(x, y, \mathbb{I}_a, \mathbb{I}_b)$, it will output $(-\infty \cdot \mathbb{I}_b, -\infty \cdot \mathbb{I}_a, -\mathbb{I}_a, -\mathbb{I}_b)$

Repeat a token once it's printed

There are going to be constructions were we would like to be able to, once the transformer prints a token, repeat it indefinitely. We can achieve this easily, making use of some of the tools that we had defined. First, we need to normalize the transformer. **i know i define normalization later on, but i think this is the place in the text where i should write this primitive.** Second, we need to flag the token with the TE. Third, we will apply a constant function to the flagged vector. Since we are trying to replicate a token that was printed, i.e. not part of the input, its vector will be the last one. This is because once the transformer prints it for the first time, we will start forcing it to repeat it. It's important that the constant function is applied in the last layers so that the vector that goes into the OUTPUT layer (which is just a projection) is the one making all the probability going to the desired token.

Notice that this only works if the token does not appear in the input word. That case can be solved using the non-uniformity of the model. If the input word is of size n , with the PE we can flag the first n vectors. Then, we can use that flag to make them 0 in the coordinate used by the TE for flagging the token to repeat. We make them 0 using a constant function as described in a previous section.

Force to halt after certain number of steps of CoT

With a similar strategy to the previous section, with the PE we can flag when a certain position is reached. Then, if the transformer is normalized, we can make the last vector what ever we want (since it will be the flagged by the PE). Particularly, we can force it to center the probability in q_{accept} or q_{reject} .

Notice that this primitive with the one before, makes us able to always assume that the transformer uses all of its budget before halting. If that wasn't the case, we can extend the alphabet with two dummy versions of the decision tokens and force the transformer to print those tokens instead of the real ones¹. Then, we can force it to repeat the one it printed until the last position of the budget. At that point, we can force it to print the real decision token corresponding to the one it was repeating (if it already printed one) or force it to choose one.

¹ This is achieved giving the probability of each decision token to its dummy version.

SF: ¿qué eran b_1 y b_2 ?

SF: third? b del a, b .

SF: de nuevo, redacción recargada. Otra cosa: ¿"indefinitamente" qué quiere decir? ¿Es una cantidad fija pero arbitraria de veces?

SF: aca, ser más preciso. Se puede dar el número de sección (si les ponés números)

SF: no entendí esta explicación

Copy one vector into another with ATTN

We will see that a useful operation will be to move values from a vector v_i to a vector v_j . This can only be done if $i < j$ since the attention mechanism does not allow for a vector to attend to other that is to the right, i.e, v_k can attend to v_l iff $k \leq l$.

We will copy values from v_i to v_j in two steps. First, we will make 0 the coordinates of v_j that will receive the values. Second, we will make v_j only attend to v_i in an attention layer. In said attention layer, we will have a $W_V = id$ and W_O such that it outputs the desire coordinates of v_i in the coordinates we want to copy them. For example if we were looking for copying the k th coordinate of v_i to the l th coordinate of v_j , W_O will have 0 in every coordinate and 1 in the row l , column k . Then the vector coming out of the ATTN layer will have 0 in every coordinate, except in the l th, where it will be $(v_i)_k$.

3.2 Normalization

For an easier handling of transformers, we define a notion of normalized transformer. This notion will specify properties of the OUTPUT matrix and of the vector consumed by this step.

Definition 3.2.1 (Normalized transformer). We will say a transformer is normalized when it meets two conditions. First, the matrix of the OUTPUT has to be a projection, i.e. it's a diagonal matrix with only ones in its diagonal. Second, the last vector after the last iteration has to only be composed of $-\infty$ in every projected coordinate except, one which has to be ∞ . The non projected coordinates are free to be any value.

SF veo que más adelante se habla de la "first condition" y la "second condition". Sería bueno numerarlas o darles in nombre para que sea más fácil referirse a ellas.

The motivation for this definition is to simplify the OUTPUT layer (first constraint) and to have more control over the probability distribution generated (second constraint). The exact values of the last vectors are such that after projecting and applying softmax in the OUTPUT layer we get a deterministic distribution with 100% of the probability centered in only one token.

SF Decí que lo que sigue es la implementación de un algoritmo para normalizar cualquier transformer.

Algorithm for normalizing

not sure about title

Let's see how to normalize any transformer. Let's take a transformer and modify it to meet the first constraint.

Every linear transformation $M : \mathbb{R}^d \rightarrow \mathbb{R}^{|\Sigma|}$ with $d \geq |\Sigma|$ can be decomposed in two steps.²

$$\begin{pmatrix} x_1 \\ \vdots \\ x_d \end{pmatrix} \xrightarrow{M'} \begin{pmatrix} (Mx)_1 \\ \vdots \\ (Mx)_{|\Sigma|} \\ v \end{pmatrix} \xrightarrow{\text{projection}} \begin{pmatrix} (Mx)_1 \\ \vdots \\ (Mx)_{|\Sigma|} \end{pmatrix}$$

² We can always add more coordinates to the embedding if $d < |\Sigma|$. Since the language is fixed, it will be a constant number of coordinates, thus not changing the class of the transformer.

SF: "first constraint" se puede cambiar por el número o nombre que le hayas dado a la primera condición en la definición. Lo mismo para la segunda.

where $v \in \mathbb{R}^{d-|\Sigma|}$ can have any values.

Let's TF be a transformer and M be the matrix of the OUTPUT layer. Using the decomposition just presented, and the strategy defined in the previous section for applying linear transformations, we can change the matrix of the OUTPUT layer to be only a projection. The linear transformation has to be applied to the last vector.

Let's now make it meet the second condition.

Once TF has been modified to meet the first condition, we can use the maximum operation described in the previous section to make all the positions of the vector 0 except the one that was the maximum before. Now, if the maximum is positive, multiplying by ∞ , then subtracting 1 to every coordinate and multiplying by ∞ again gets us what we wanted.

SF: acá podés ser más preciso, si ponés números a secciones

$$\begin{pmatrix} 0 \\ \vdots \\ 0 \\ x_{max} \\ 0 \\ \vdots \\ 0 \end{pmatrix} \xrightarrow[\infty]{\text{multiplying by}} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ \infty \\ 0 \\ \vdots \\ 0 \end{pmatrix} \xrightarrow{\text{subtracting 1}} \begin{pmatrix} -1 \\ \vdots \\ -1 \\ \infty - 1 \\ -1 \\ \vdots \\ -1 \end{pmatrix} \xrightarrow[\infty]{\text{multiplying by}} \begin{pmatrix} -\infty \\ \vdots \\ -\infty \\ \infty \\ -\infty \\ \vdots \\ -\infty \end{pmatrix}$$

It could be the case that the maximum was negative, in that case the resulting vector won't be what we are looking for. Let's see how to make the maximum strictly positive so we can add this mechanism in the middle.

By this point our vector looks like $(0, \dots, 0, x_{max}, 0, \dots, 0)$, we are going to transform it to $(0, \dots, 0, |x_{max}|, 0, \dots, 0)$. Remember that the coordinates we are interested in are only the first $|\Sigma|$. Let's call v to the block containing these coordinates. Our algorithm will do the following:

$$\begin{pmatrix} v \\ \vdots \end{pmatrix} \rightarrow \begin{pmatrix} v \\ -v \\ \vdots \end{pmatrix} \rightarrow \begin{pmatrix} \text{relu}(v) \\ \text{relu}(-v) \\ \vdots \end{pmatrix} \rightarrow \begin{pmatrix} \text{relu}(v) + \text{relu}(-v) \\ \vdots \end{pmatrix}$$

where relu is applied position wise. This works since for all x it holds that $\text{relu}(x) + \text{relu}(-x) = |x|$.

The first step can be done in the same way as the first half of the linear transformations. For the second step we can take max coordinate in v and $-v$. This works because all the coordinates are 0 except one, which if it is negative will become 0 and if it is non-negative will remain the same. For adding both vectors we can use one FF with the following parameters. Notice that this works since all the coordinates of $\text{relu}(-v)$ are non-negative.

$$W_1 = id^{d \times d} \quad W_2 = \begin{pmatrix} 0^{|\Sigma| \times |\Sigma|} & id^{|\Sigma| \times |\Sigma|} & 0^{|\Sigma| \times (d-2|\Sigma|)} \\ 0^{d-|\Sigma| \times |\Sigma|} & 0^{d-|\Sigma| \times |\Sigma|} & 0^{d-|\Sigma| \times d-2|\Sigma|} \end{pmatrix} \quad b_1 = b_2 = (0^{d \times d})$$

3.3 Boolean Operations

The main objective of this chapter was to prove that the $\text{CoT}[T(n), d(n)]$ classes are closed

SF: section. Además, usar presente, no pasado

under boolean operations.

Let's show that the complement of a language in $\text{CoT}[T(n), d(n)]$ is in $\text{CoT}[T(n), d(n)]$ and that the disjunction of two language of the same class, stays in the same class.

Complement of a language

Let's $\mathcal{L} \in \text{CoT}[T(n), d(n)]$, then there is a family of transformers $\{\text{TF}_n\}_{n \in \mathbb{N}}$ of embedding size $d(n)$ that after $T(n)$ steps of CoT prints q_{accept} or q_{reject} depending on if the input word is in the language. For each of the transformers in the family we can define one that behaves exactly the same (internally and externally) but prints the flipped token.

Let's TF be a transformer, we will construct TF' that flips the decision token of TF , i.e if $\text{TF}^*(\alpha) = q_{\text{accept}}$, then $\text{TF}'^*(\alpha) = q_{\text{reject}}$ and if $\text{TF}^*(\alpha) = q_{\text{reject}}$, then $\text{TF}'^*(\alpha) = q_{\text{accept}}$.

This can be easily done swapping the rows associated to q_{accept} and q_{reject} in the matrix of the **OUTPUT** layer of TF . Thus, when the largest probability is associated to q_{accept} in TF , that same number will be associated to q_{reject} in TF' .

Formally, TF' will be exactly the same as TF except in the **OUTPUT** layer, where the matrix will be different. If M is the matrix of the **OUTPUT** layer of TF , i is the position of the token q_{accept} and j the one of q_{reject} , then the matrix of the **OUTPUT** layer of TF' will be $M' = PM$ where P is the matrix that flips the rows i and j .

$$M'_{kl} = \begin{cases} 1 & \text{if } k = l, k \neq i \text{ and } k \neq j \\ 1 & \text{if } k = j \text{ and } l = i \\ 1 & \text{if } k = i \text{ and } l = j \\ 0 & \text{otherwise} \end{cases}$$

TF' has the same budget and embedding size that TF .

We can define the family $\{\text{TF}'_n\}_{n \in \mathbb{N}}$ where TF'_i is the negation of TF_i . Then $\{\text{TF}'_n\}_{n \in \mathbb{N}}$ computes the complement of $\{\text{TF}_n\}_{n \in \mathbb{N}}$. Since for every $i \in \mathbb{N}$, TF'_i has the same budget and embedding size that TF_i , $\mathcal{L}$ is in $\text{CoT}[T(n), d(n)]$.

Disjunction of two languages

Let's $\mathcal{L}_1, \mathcal{L}_2 \in \text{CoT}[T(n), d(n)]$, then there are is a family of transformers $\{\text{TF1}_n\}_{n \in \mathbb{N}}$ (resp. $\{\text{TF2}_n\}_{n \in \mathbb{N}}$) of embedding size $d_1(n)$ (resp. $d_2(n)$) $\in O(d(n))$ that after $T_1(n)$ (resp. $T_2(n)$) $\in T(n)$ steps of CoT that computes $\mathcal{L}_1$ (resp. $\mathcal{L}_2$).

We will construct a family $\{\text{TF}_n\}_{n \in \mathbb{N}}$ that will compute $\mathcal{L}_1 \cup \mathcal{L}_2$. It will do it in $T_3(n) = T_1(n) + T_2(n) + 1 \in O(T(n))$ steps of CoT and it will have embedding size $d_3(n) \in O(d(n))$.

The idea behind the construction will be to run first one of the transformers, then run the other and finally compare both results. Let's assume that when the input is α , the tape of the transformer TF1 ends up looking like $\alpha\sigma q1$, where $\sigma \in \Sigma^*$ and $q1$ is a decision symbol of TF1 . In the same way, let's call $\alpha\tau q2$ to the content of the tape of TF2 when it halts after being input with α . The tape of the transformer computing the disjunction of TF1 and TF2 , at the end of the computation will look like $\alpha\sigma q1\tau q2 q3$, where σ and τ are the intermediate tokens generated by TF1 and TF2 , $q1$ and $q2$ are their acceptance or rejection tokens and $q3$ is the disjunction of $q1$ and $q2$.

Let's construct TF3 from TF1 and TF2 . Since we can't change the body of the transformer during the iterations we will require to use some tricks.

SF: Lo malo de usar $q1$ o $q2$ sin subíndices es que cuando los concatenás, se confunde con dos palabras.

We grow the dimension of the embedding for running the body of both transformers at the same time over the same vectors but independently. In the final layers we will choose which probability distribution will go to the output layer, this will make TF3 print the token generated by TF1 or TF2. As always, we will assume that the transformers are normalized.

We will use the first half of the embedding for the transformer TF1 and the second for the transformer TF2. We will have extra coordinates for flags at the end, but we will ignore them by now.

$$\text{TE3}(\sigma)_i = \begin{cases} \text{TE1}(\sigma)_i & \text{if } i \leq d_1(|\alpha|) \\ \text{TE2}(\sigma)_{i-d_1(|\alpha|)} & \text{if } d_1(|\alpha|) < i \leq d_2(|\alpha|) \\ 0 & \text{if } d_2(|\alpha|) < i \end{cases}$$

$$\text{PE3}(n)_i = \begin{cases} \text{PE1}(n)_i & \text{if } i \leq d_1(|\alpha|) \\ \text{PE2}(n)_{i-d_1(|\alpha|)} & \text{if } d_1(|\alpha|) < i \leq d_2(|\alpha|) \\ \text{flags} & \text{if } d_2(|\alpha|) < i \end{cases}$$

there is a problem in this definition, PE2 is defined over $[|\alpha| + T2(|\alpha|)]$ and here we are using it over $[|\alpha| + T2(|\alpha|) + T1(|\alpha|)]$. This is fixed later on but im not sure if i want to note it right now.

The transformer TF3 will have one layer for each layer of TF1 and each of TF2. The first layers, will correspond to the body of TF1. They will be constructed ignoring the half of the embedding where the coordinates coming from TF2 are. The second half will be the other way around but with the layers of TF2 and ignoring TF1 coordinates. Then, all the matrix in the layers of the first part of the body of TF3 will look like $W3_1$ and the ones in the second layers will look like $W3_2$ where $W1$ and $W2$ are the matrix of the corresponding layers in TF1 and TF2.

$$W3_1 = \begin{pmatrix} W1 & 0 \\ 0 & 0 \end{pmatrix} \qquad W3_2 = \begin{pmatrix} 0 & 0 \\ 0 & W2 \end{pmatrix}$$

SF: matrices con nombres no estándar y confusos

It is important to notice that the values of the coordinates corresponding to one transformer does not affect the values of the other. Thus, if the tape of the transformer has content β , then at the end of the body of TF3, the last vector would look like (x, y, flags) where x is the last vector at the end of the body of TF1 when its tape content is β , and y the one at the end of TF2 when its tape content is β . Therefore, if we always choose to transform said vector into $(x, 0, \dots, 0)$ then TF3 will produce the same tape as TF1. This works for generating the first $T1(|\alpha|)$ tokens which are $\sigma q1$. Thus, we would like to apply this transformation only to the vectors generating them. Therefore, we need to flag them with the position embedding. Since $v_{|\alpha|+i}$ carries the distribution to output in the $i + 1$ th step, i.e the $|\alpha| + i + 1$ th token of the tape, the vectors that we need to flag will be $v_{|\alpha|}, v_{|\alpha|+1}, \dots, v_{|\alpha|+T1(|\alpha|)-1}$.

After running $T1(|\alpha|)$ steps of this construction, the tape will look like $\alpha\sigma q1$. We want to now generate τ . Just changing what we choose before the OUTPUT layer does not work since what the construction will produce would be $\text{TF2}(\alpha\sigma q1)$, and we are looking for $\text{TF2}(\alpha)$.

Let's see how to run TF2 in this construction ignoring the tokens produced by TF1. If we are able to make those tokens invisible to TF2, we are done. We need for TF2 to not attend to them (this can be done using the primitive presented in the previous section) and for its position encoding to ignore them, therefore we need to "shift" it. Let's see first how to fix the position encoding of TF3.

$$\text{PE3}(n)_i = \begin{cases} \text{PE1}(n)_i & \text{if } i \leq d_1(|\alpha|) \\ \text{shiftedPE2}(n)_{i-d_1(|\alpha|)} & \text{if } d_1(|\alpha|) < i \leq d_2(|\alpha|) \\ \text{flags} & \text{if } d_2(|\alpha|) < i \end{cases}$$

$$\text{shiftedPE2}(n) = \begin{cases} \text{PE2}(n) & \text{if } n \leq |\alpha| \\ 0 & \text{if } |\alpha| < n < T1(|\alpha|) \\ \text{PE2}(n - T1(|\alpha|)) & \text{if } T1(|\alpha|) < n \end{cases}$$

The first case is for the positions occupied by the input, we don't want to change them. The second case is for the positions occupied by the tokens generated by TF1, we don't care about them. The third case is for the tokens that TF2 will generate, we want to make it believe that they are in their natural positions, i.e shifted $T1(|\alpha|)$ positions left (the positions occupied by the generated for TF1). We will also need to change the flag to the vectors after the number $\alpha + T1(|\alpha|)$ to choose the distribution generated by TF2 instead of TF1.

With the construction like this we can almost produce the tape of TF1 followed by the tape of TF2 except for one token. We need to do something especial for the first token generated by TF2. The distribution corresponding to it, it's in the vector $v_{|\alpha|}$ but the last vector of the first run of TF2 is $v_{|\alpha|+T1(|\alpha|)}$, the one coming from the last token of TF1. Therefore, before choosing the distribution, we need to copy the vector. This can be done with the primitives of the previous section.

With this last fix, we are able to generate the tape of TF1 followed by the tape of TF2. We are only missing how to compute q_3 . This can be easily done making $v_{|\alpha|+T1(|\alpha|)+T2(\alpha)}$ only attend to itself and to $v_{|\alpha|+T1(|\alpha|)}$, i.e. to the vectors coming from q_1 and q_2 .

We will modify the TE3 function to add a flag in the last coordinate (let's call it f_c) for the decision tokens of TF1 and TF2.

$$\text{TE3}(\sigma)_{f_c} = \begin{cases} 1 & \text{if } \sigma = q_{\text{accept}} \vee \sigma = q_{\text{accept}} \\ 0 & \text{otherwise} \end{cases}$$

Then if we add $v_{|\alpha|+T1(|\alpha|)}$ to $v_{|\alpha|+T1(|\alpha|)+T2(\alpha)}$ we will have 1 or 2 in the coordinate f_c iff TF1 or TF2 accepted α . Now we can make all of $v_{|\alpha|+T1(|\alpha|)+T2(\alpha)}$ 0, except the f_c coordinate. Before going to the output layer we need to move the value of f_c to the coordinate that will project to the probability of q_{accept} and add some number larger than 0 and smaller than 1 to the coordinate that will define the probability of q_{reject} .

Therefore, the token with the maximum probability will be q_{accept} iff TF1 or TF2 accepted α , and it will be q_{reject} in the other case. Then, TF3 computes the union of the languages of TF1 and TF2. The number of steps of CoT that it needs is the same as TF1 plus TF2 plus 1 extra for taking the disjunction, thus in $O(\cdot)$ notation it takes the same as the maximum between TF1 and TF2. The embedding dimension of TF3 is the same as the dimension of TF1 plus TF2 plus some constant number of flags, thus in $O(\cdot)$ notation it's the same as the maximum between TF1 and TF2.

Therefore, the language computed by the family of transformers $\{\text{TF3}_n\}_{n \in \mathbb{N}}$ is in $\text{CoT}[T(n), d(n)]$. Thus, $\mathcal{L1} \cup \mathcal{L2} \in \text{CoT}[T(n), d(n)]$.

4. MAIN RESULTS

5. CONCLUSIONS